In 3D: partition a box into small boxes of volume $\Delta x\, \Delta y\, \Delta z$:

$$\iiint_\Omega 1\, dx\, dy\, dz = \text{volume of } \Omega.$$

For a general region $D \subseteq \mathbb{R}^2$:

$$\iint_D dx\, dy = \text{area of } D, \qquad \iiint_\Omega dx\, dy\, dz = \text{volume of } \Omega.$$

2.2 Computing Areas and Volumes in Cartesian Coordinates

For a region $D = \{(x, y) : a \le x \le b,\ g(x) \le y \le h(x)\}$:

$$\iint_D dx\, dy = \int_a^b \int_{g(x)}^{h(x)} dy\, dx = \int_a^b \big(h(x) - g(x)\big)\, dx.$$

This is the formula for the area between two curves — the same formula from first-year calculus, now interpreted as a double integral.

Example 1: area of a triangle. $D = \{(x, y) : 0 \le x \le 1,\ 0 \le y \le x\}$:

$$\iint_D dx\, dy = \int_0^1 \int_0^x dy\, dx = \int_0^1 x\, dx = \tfrac{1}{2}. \quad \checkmark$$

Example 2: volume of a pyramid. $\Omega = \{(x, y, z) : x, y, z \ge 0,\ x + y + z \le 1\}$:

$$\iiint_\Omega dx\, dy\, dz = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} dz\, dy\, dx = \int_0^1 \int_0^{1-x} (1 - x - y)\, dy\, dx = \int_0^1 \tfrac{(1-x)^2}{2}\, dx = \tfrac{1}{6}.$$

2.3 The Integral of $f(x, y)$ as Volume Under a Surface

If $f(x, y) \ge 0$, then $\iint_D f(x, y)\, dx\, dy$ is the **volume of the solid** lying between the region D in the xy-plane and the surface $z = f(x, y)$ above it.

This follows from the Riemann sum interpretation: each term $f(x_i, y_j)\, \Delta x\, \Delta y$ is the volume of a thin rectangular column of height $f(x_i, y_j)$ and base area $\Delta x\, \Delta y$.

Example 3: volume under a paraboloid. $f(x, y) = 1 - x^2 - y^2$, $D = \{(x, y) : x^2 + y^2 \le 1\}$:

In Cartesian coordinates this is hard to set up. We will return to it using polar coordinates in Part 3. For now, note that $\iint_D (1 - x^2 - y^2)\, dx\, dy$ should give $\pi/2$ (by symmetry: the paraboloid reaches height 1 at the origin and 0 on the unit circle).

Example 4: probability (chemistry motivation). The Maxwell–Boltzmann speed distribution in 2D is $f(v_x, v_y) \propto e^{-(v_x^2 + v_y^2)/2k_B T/m}$. The probability of finding the speed in a disk of radius v_0 is:

$$P(|\mathbf{v}| \le v_0) = C \iint_{v_x^2 + v_y^2 \le v_0^2} e^{-(v_x^2 + v_y^2)/2k_B T/m}\, dv_x\, dv_y.$$

This integral is essentially the Gaussian integral in 2D, which we will evaluate exactly using polar coordinates and the famous trick of Meeting 4.

2.4 The Area of a Disk: Why We Need New Coordinates

Let $D = \{(x, y) : x^2 + y^2 \leq R^2\}$ be the disk of radius R. In Cartesian coordinates:

$$\iint_D dx\, dy = \int_{-R}^{R} \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} dy\, dx = \int_{-R}^{R} 2\sqrt{R^2 - x^2}\, dx.$$

This requires a trigonometric substitution and is laborious. The answer is, of course, πR^2, but the Cartesian route gives no insight into *why*.

In polar coordinates, the disk is simply $\{0 \leq r \leq R,\ 0 \leq \theta \leq 2\pi\}$ — a rectangle in (r, θ)-space. To use this, we need to know how areas transform under the coordinate change. The answer involves the Jacobian determinant, and we will derive it now.

Meeting 4

Part 3 — Change of Variables and Fubini's Theorem (~50 min)

3.1 The Change of Variables Formula

We hinted in Part 1 that the Jacobian determinant is the key to coordinate changes. Here is the precise statement.

Theorem (Change of Variables)

Let $\Phi \colon U \to V$ be a smooth bijection between open sets in $\mathbb{R}^n$ with $\det D\Phi \neq 0$ on U. Then for any integrable function $f \colon V \to \mathbb{R}$:

$$\int_V f(x)\, dx = \int_U f(\Phi(u))\, |\det D\Phi(u)|\, du.$$

Why $|\det D\Phi|$? A small box $[u_0, u_0 + du_1] \times \cdots$ in u-space is sent by Φ to an approximate parallelotope in x-space spanned by the columns $(\partial\Phi/\partial u_i)\, du_i$. The volume of this parallelotope is $|\det D\Phi|\, du_1 \cdots du_n$. This is the geometric meaning of the Jacobian determinant: it measures how much Φ stretches or compresses volumes. At points where $|\det D\Phi| = 0$, the map collapses volume — which is exactly why coordinates degenerate there.

Area of a disk. With $\Phi(r, \theta) = (r \cos\theta, r \sin\theta)$, $|\det D\Phi| = r$:

$$\iint_D dx\, dy = \int_0^{2\pi} \int_0^R r\, dr\, d\theta = 2\pi \cdot \frac{R^2}{2} = \pi R^2. \quad \checkmark$$

The factor r converts the polar "rectangle" $dr\, d\theta$ into the true area element in the plane.

Volume of a ball. With spherical coordinates, $|\det D\Phi| = r^2 \sin\theta$:

$$\iiint_{|\mathbf{x}| \leq R} dx\, dy\, dz = \int_0^{2\pi} \int_0^{\pi} \int_0^R r^2 \sin\theta\, dr\, d\theta\, d\varphi = 2\pi \cdot 2 \cdot \frac{R^3}{3} = \frac{4\pi R^3}{3}. \quad \checkmark$$

3.2 Fubini's Theorem

Theorem (Fubini)

If $f\colon [a,b]\times[c,d]\to\mathbb{R}$ is continuous, then:

$$\iint_{[a,b]\times[c,d]} f(x,y)\,dx\,dy = \int_a^b\int_c^d f(x,y)\,dy\,dx = \int_c^d\int_a^b f(x,y)\,dx\,dy.$$

The order of integration can be freely exchanged.

Fubini's theorem extends to general regions (with appropriate hypotheses on f). Changing the order of integration can dramatically simplify a computation.

Example. $\displaystyle\int_0^1\int_x^1 e^{y^2}\,dy\,dx.$

The inner integral $\int_x^1 e^{y^2}\,dy$ cannot be computed in closed form. Switching order: the region is $\{0\le x\le y,\ 0\le y\le 1\}$, so:

$$\int_0^1\int_0^y e^{y^2}\,dx\,dy = \int_0^1 y\,e^{y^2}\,dy = \left[\tfrac{1}{2}e^{y^2}\right]_0^1 = \tfrac{1}{2}(e-1).$$

3.3 Exercises

Exercise 5. Compute the following integrals using polar coordinates.

(a) $\displaystyle\iint_D e^{-(x^2+y^2)}\,dx\,dy$ where $D=\{x^2+y^2\le R^2\}$.

(b) $\displaystyle\iint_D \sqrt{x^2+y^2}\,dx\,dy$ where D is the unit disk.

(c) $\displaystyle\iint_D (x^2+y^2)\,dx\,dy$ where $D=\{1\le x^2+y^2\le 4\}$ (annulus).

Answers: (a) $\pi(1-e^{-R^2})$; *(b)* $2\pi/3$; *(c)* $\pi\int_1^2 r^2\cdot r\,dr\cdot 2\pi$ $\to$ *wait,* $\int_0^{2\pi}d\theta\int_1^2 r^2\cdot r\,dr = 2\pi\cdot[r^4/4]_1^2 = 2\pi\cdot 15/4 = 15\pi/2$.

Exercise 6. Compute the volume under the paraboloid $z=1-x^2-y^2$ above the unit disk $D=\{x^2+y^2\le 1\}$.

Use polar coordinates: $\int_0^{2\pi}\int_0^1(1-r^2)r\,dr\,d\theta = 2\pi\int_0^1(r-r^3)\,dr = 2\pi\cdot[r^2/2-r^4/4]_0^1 = 2\pi\cdot 1/4 = \pi/2$.

Exercise 7. Compute $\displaystyle\iint_D xy\,dx\,dy$ where $D=\{(x,y): x^2+y^2\le 1,\ x\ge 0,\ y\ge 0\}$ (first quadrant of unit disk).

In polar: $\int_0^{\pi/2}\int_0^1 r\cos\theta\cdot r\sin\theta\cdot r\,dr\,d\theta = \int_0^{\pi/2}\sin\theta\cos\theta\,d\theta\cdot\int_0^1 r^3\,dr = \frac{1}{2}\cdot\frac{1}{4} = \frac{1}{8}$.

Exercise 8. Switch the order of integration and evaluate:

$$\int_0^1\int_y^1 \sin(\pi x^2)\,dx\,dy.$$

Region: $0\le y\le x\le 1$. *Switching:* $\int_0^1\int_0^x \sin(\pi x^2)\,dy\,dx = \int_0^1 x\sin(\pi x^2)\,dx = [-\cos(\pi x^2)/(2\pi)]_0^1 = 1/(\pi)$.

Exercise 9 (chemistry). The electron density of the hydrogen 1s orbital is $\rho(r) = \frac{1}{\pi a_0^3} e^{-2r/a_0}$. Verify the normalisation $\int_0^\infty \int_0^\pi \int_0^{2\pi} \rho(r)\, r^2 \sin\theta\, d\varphi\, d\theta\, dr = 1$.

Angular integral: 4π. *Radial:* $\frac{4}{a_0^3} \int_0^\infty r^2 e^{-2r/a_0}\, dr = \frac{4}{a_0^3} \cdot 2! \cdot (a_0/2)^3 = \frac{4}{a_0^3} \cdot \frac{a_0^3}{4} = 1$.

Part 4 — Classic Results (~40 min)

4.1 The Gaussian Integral

$$\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}.$$

Proof via double integral and polar coordinates. Let $I = \int_{-\infty}^{\infty} e^{-x^2}\, dx$. Consider I^2:

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2}\, dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2}\, dy\right) = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\, dx\, dy.$$

Converting to polar coordinates ($r \in [0,\infty)$, $\theta \in [0, 2\pi)$):

$$I^2 = \int_0^{2\pi} \int_0^\infty e^{-r^2}\, r\, dr\, d\theta = 2\pi \int_0^\infty r\, e^{-r^2}\, dr = 2\pi \cdot \left[-\tfrac{1}{2} e^{-r^2}\right]_0^\infty = 2\pi \cdot \tfrac{1}{2} = \pi.$$

Therefore $I = \sqrt{\pi}$. $\square$

Why this matters in chemistry. The normalised Gaussian $e^{-x^2/2}/\sqrt{2\pi}$ is the probability density of the normal distribution. Every measurement uncertainty is modelled by a Gaussian; every spectral line has a Gaussian profile. The factor $\sqrt{\pi}$ that appears throughout quantum chemistry — in the normalisation of harmonic oscillator eigenfunctions, in the error function, in the Debye–Waller factor — originates here.

4.2 The Basel Problem: $\zeta(2) = \pi^2/6$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{4} + \frac{1}{9} + \cdots = \frac{\pi^2}{6}.$$

This was proved by Euler in 1734 and shocked the mathematical community. Here is a proof using double integrals.

Step 1. Use the geometric series: for $0 \le xy < 1$, $\frac{1}{1-xy} = \sum_{n=0}^\infty (xy)^n$. Integrating over the unit square $(0,1)^2$:

$$\iint_{(0,1)^2} \frac{dx\, dy}{1-xy} = \sum_{n=0}^\infty \iint_{(0,1)^2} (xy)^n\, dx\, dy = \sum_{n=0}^\infty \frac{1}{(n+1)^2} = \sum_{n=1}^\infty \frac{1}{n^2}.$$

Step 2: change of variables. Apply the 45° rotation $u = \dfrac{x+y}{\sqrt{2}}$, $v = \dfrac{x-y}{\sqrt{2}}$, with inverse $x = \dfrac{u+v}{\sqrt{2}}$, $y = \dfrac{u-v}{\sqrt{2}}$. The Jacobian of a rotation is 1, so $dx\, dy = du\, dv$. Also:

$$xy = \frac{(u+v)(u-v)}{2} = \frac{u^2-v^2}{2}, \qquad 1 - xy = 1 - \frac{u^2-v^2}{2} = \frac{2-u^2+v^2}{2}.$$

The unit square $(0,1)^2$ in (x,y) maps to a square rotated $45°$ in (u,v), with vertices at $(0,0) \to (0,0)$, $(1,0) \to (1/\sqrt{2}, -1/\sqrt{2})$, $(0,1) \to (1/\sqrt{2}, 1/\sqrt{2})$, $(1,1) \to (\sqrt{2}, 0)$. By the symmetry $v \mapsto -v$ (which leaves the integrand unchanged), the integral over the full rotated square equals twice the integral over the upper half, which splits into two triangles:

$$T_1 = \left\{0 \le v \le u \le \tfrac{1}{\sqrt{2}}\right\}, \qquad T_2 = \left\{0 \le v \le \tfrac{1}{\sqrt{2}} - u,\ \tfrac{1}{\sqrt{2}} \le u \le \sqrt{2}\right\}.$$

By the further symmetry $u \mapsto \sqrt{2} - u$ of T_2 relative to T_1, both triangles give the same integral, so:

$$\iint_{(0,1)^2} \frac{dx\,dy}{1-xy} = 4 \iint_{T_1} \frac{du\,dv}{1 - \frac{u^2-v^2}{2}} = 4 \int_0^{1/\sqrt{2}} \int_0^u \frac{2\,dv\,du}{2 - u^2 + v^2}.$$

Step 3: the inner integral. For fixed u, integrate over $v \in [0,u]$:

$$\int_0^u \frac{2\,dv}{2 - u^2 + v^2}.$$

Write $a^2 = 2 - u^2$ (valid for $u < \sqrt{2}$, which holds on T_1 since $u \le 1/\sqrt{2}$). Using $\int dv/(a^2 + v^2) = \frac{1}{a}\arctan(v/a) + C$:

$$\int_0^u \frac{2\,dv}{a^2+v^2} = \frac{2}{a}\arctan\!\left(\frac{v}{a}\right)\Bigg|_0^u = \frac{2}{\sqrt{2-u^2}}\arctan\!\left(\frac{u}{\sqrt{2-u^2}}\right).$$

Step 4: recognise the arctan. Substitute $u = \sqrt{2}\sin\phi$, so $\sqrt{2-u^2} = \sqrt{2}\cos\phi$ and $u/\sqrt{2-u^2} = \tan\phi$. Thus $\arctan(u/\sqrt{2-u^2}) = \phi = \arcsin(u/\sqrt{2})$. At $u = 0$: $\phi = 0$. At $u = 1/\sqrt{2}$: $\phi = \pi/4$.

The inner integral becomes $\dfrac{2}{\sqrt{2-u^2}}\arcsin\!\left(\dfrac{u}{\sqrt{2}}\right)$.

Step 5: the outer integral.

$$\frac{\pi^2}{6} \stackrel{?}{=} 4\int_0^{1/\sqrt{2}} \frac{2}{\sqrt{2-u^2}}\arcsin\!\left(\frac{u}{\sqrt{2}}\right) du.$$

Substitute $u = \sqrt{2}\sin\phi$, $du = \sqrt{2}\cos\phi\,d\phi$, $\sqrt{2-u^2} = \sqrt{2}\cos\phi$; limits $0 \to 0$, $1/\sqrt{2} \to \pi/4$:

$$4\int_0^{\pi/4} \frac{2}{\sqrt{2}\cos\phi}\cdot\phi\cdot\sqrt{2}\cos\phi\,d\phi = 4\int_0^{\pi/4} 2\phi\,d\phi = 8\left[\frac{\phi^2}{2}\right]_0^{\pi/4} = 4\cdot\frac{\pi^2}{16} = \frac{\pi^2}{4}.$$

Wait — this gives $\pi^2/4$, not $\pi^2/6$. Recount: the symmetry argument above contributes a factor of 4 (factor 2 for $v \mapsto -v$, factor 2 for the two triangles T_1 and T_2), but we also have the factor of 2 from $\frac{2\,dv}{2-u^2+v^2}$, giving overall factor 8 on T_1. Let us redo the splitting carefully.

The rotated square has vertices $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$, $(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$, $(\sqrt{2}, 0)$, $(0,0)$. It consists of:

$$S^+ = \{0 \le v \le u,\ u + v \le \sqrt{2}\}, \quad S^- = \{-u \le v \le 0,\ u - v \le \sqrt{2}\}.$$

By $v \mapsto -v$ symmetry, $\iint_{S^+} = \iint_{S^-}$, so:

$$\iint_{(0,1)^2} \frac{dx\,dy}{1-xy} = 2\iint_{S^+} \frac{2\,du\,dv}{2 - u^2 + v^2}.$$

The region S^+ further splits at $u = 1/\sqrt{2}$:

$$T_1 = \{0 \le v \le u \le \tfrac{1}{\sqrt{2}}\}, \quad T_2 = \{0 \le v \le \sqrt{2} - u,\ \tfrac{1}{\sqrt{2}} \le u \le \sqrt{2}\}.$$

By the substitution $u \mapsto \sqrt{2} - u$, $v \mapsto v$ (which maps T_2 to T_1 and preserves $2 - u^2 + v^2 \mapsto 2 - (\sqrt{2} - u)^2 + v^2 = u^2 + 2\sqrt{2}u - v^2$) — actually T_1 and T_2 are *not* related by this simple symmetry. We integrate them separately.

On T_1: (as computed above) $2\int_0^{1/\sqrt{2}} \frac{2}{\sqrt{2-u^2}} \arcsin\left(\frac{u}{\sqrt{2}}\right) du = 2\int_0^{\pi/4} 2\phi\, d\phi = 4 \cdot \frac{\pi^2}{32} = \frac{\pi^2}{8}$.

On T_2: By the substitution $u = \sqrt{2} - s$, $v = v$ (so s runs $0 \to 1/\sqrt{2}$, $v \in [0, s]$):

$$2 - u^2 + v^2 = 2 - (\sqrt{2} - s)^2 + v^2 = 2s\sqrt{2} - s^2 + v^2.$$

The inner v-integral: $\displaystyle\int_0^s \frac{2\, dv}{2\sqrt{2}s - s^2 + v^2}$.

Writing $b^2 = s(2\sqrt{2} - s)$: $\displaystyle\frac{2}{b}\arctan\left(\frac{v}{b}\right)\Big|_0^s = \frac{2}{b}\arctan\left(\frac{s}{b}\right)$.

Now $s/b = s/\sqrt{s(2\sqrt{2}-s)} = \sqrt{s/(2\sqrt{2}-s)}$. With $s = \sqrt{2}\sin^2\psi$ one finds $\arctan(s/b) = \psi$ and after substitution the outer integral over s yields $\pi^2/8$ as well.

Conclusion.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \iint_{(0,1)^2} \frac{dx\, dy}{1 - xy} = 2\left(\frac{\pi^2}{8} + \frac{\pi^2}{8}\right) = \frac{\pi^2}{4} \cdot \frac{1}{1}$$

Hmm. Let us restate the calculation without error. The correct accounting is:

$$\iint_{(0,1)^2} \frac{dx\, dy}{1 - xy} = \iint_{S^+ \cup S^-} \frac{2\, du\, dv}{2 - u^2 + v^2} = 2\iint_{S^+} \frac{2\, du\, dv}{2 - u^2 + v^2} = 2\left(I_{T_1} + I_{T_2}\right).$$

where

$$I_{T_1} = \int_0^{1/\sqrt{2}} \frac{2}{\sqrt{2-u^2}} \arctan\left(\frac{u}{\sqrt{2-u^2}}\right) du.$$

With $u = \sqrt{2}\sin\phi$: $\arctan(\tan\phi) = \phi$, $\sqrt{2-u^2} = \sqrt{2}\cos\phi$, $du = \sqrt{2}\cos\phi\, d\phi$:

$$I_{T_1} = \int_0^{\pi/4} \frac{2}{\sqrt{2}\cos\phi} \cdot \phi \cdot \sqrt{2}\cos\phi\, d\phi = 2\int_0^{\pi/4} \phi\, d\phi = 2 \cdot \frac{\pi^2}{32} = \frac{\pi^2}{16}.$$

By the symmetry of the integrand under $u \mapsto \sqrt{2} - u$ (which maps T_2 to T_1) and the substitution computation above, $I_{T_2} = I_{T_1} = \pi^2/16$. Therefore:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 2\left(\frac{\pi^2}{16} + \frac{\pi^2}{16}\right) = 2 \cdot \frac{\pi^2}{8} = \frac{\pi^2}{4}.$$

This gives $\pi^2/4$, not $\pi^2/6$. The error is in Step 1:

$$\iint_{(0,1)^2} (xy)^n\, dx\, dy = \left(\int_0^1 x^n dx\right)^2 = \frac{1}{(n+1)^2},$$

so $\sum_{n=0}^{\infty} \frac{1}{(n+1)^2} = \sum_{n=1}^{\infty} \frac{1}{n^2}$ is correct. The error must be in the rotation.

Let us recheck the symmetry: the full rotated domain is the square with vertices $(0,0)$, $(1/\sqrt{2},-1/\sqrt{2})$, $(\sqrt{2},0)$, $(1/\sqrt{2},1/\sqrt{2})$. In $v\mapsto -v$ symmetry the two halves are S^+ ($v\geq 0$) and S^- ($v\leq 0$), and the substitution gives:

$$\iint \frac{2\,du\,dv}{2-u^2+v^2}$$

over the whole square. The square is not $S^+\cup S^-$ with the previous description; let us directly use:

$$\begin{aligned} I &= 4\int_0^{1/\sqrt{2}}\int_0^u \frac{2\,dv\,du}{2-u^2+v^2} + 4\int_{1/\sqrt{2}}^{\sqrt{2}}\int_0^{\sqrt{2}-u} \frac{2\,dv\,du}{2-u^2+v^2} \\ &= 4(I_{T_1}+I_{T_2}). \end{aligned}$$

(Factor 4: factor 2 for $v\mapsto -v$ symmetry, times 2 for the factor in the numerator $2\,dv$.) With $I_{T_1}=\pi^2/16$ as above and $I_{T_2}=\pi^2/16$:

$$I = 4\left(\frac{\pi^2}{16}+\frac{\pi^2}{16}\right) = \frac{\pi^2}{4}.$$

This contradicts the known result. The resolution: $I_{T_2}\neq I_{T_1}$. Let us recompute I_{T_2} carefully. On T_2: $u\in[1/\sqrt{2},\sqrt{2}]$, $v\in[0,\sqrt{2}-u]$. Note $2-u^2$ is *negative* for $u>\sqrt{2}$, but on T_2 we have $u\leq\sqrt{2}$ so $2-u^2\geq 0$. At $u=1/\sqrt{2}$: $2-u^2=3/2>0$. At $u=\sqrt{2}$: $2-u^2=0$.

Inner integral: $\displaystyle\int_0^{\sqrt{2}-u}\frac{2\,dv}{(2-u^2)+v^2} = \frac{2}{\sqrt{2-u^2}}\arctan\left(\frac{\sqrt{2}-u}{\sqrt{2-u^2}}\right).$

Note $\displaystyle\frac{\sqrt{2}-u}{\sqrt{2-u^2}} = \frac{\sqrt{2}-u}{\sqrt{(\sqrt{2}-u)(\sqrt{2}+u)}} = \sqrt{\frac{\sqrt{2}-u}{\sqrt{2}+u}}.$

With $u=\sqrt{2}\cos(2\alpha)$, $\alpha\in[0,\pi/4]$: $\sqrt{2}-u=\sqrt{2}(1-\cos 2\alpha)=2\sqrt{2}\sin^2\alpha$, $\sqrt{2}+u=\sqrt{2}(1+\cos 2\alpha)=2\sqrt{2}\cos^2\alpha$, $2-u^2=2\sin(2\alpha)\sqrt{2}\cdot\ldots$ — this substitution becomes complicated.

A cleaner approach uses $u=\sqrt{2}\sin\phi$ with $\phi\in[\pi/4,\pi/2]$ for T_2: $\sqrt{2}-u=\sqrt{2}(1-\sin\phi)$, $\sqrt{2-u^2}=\sqrt{2}\cos\phi$, $\displaystyle\frac{\sqrt{2}-u}{\sqrt{2-u^2}}=\frac{1-\sin\phi}{\cos\phi}=\tan(\pi/4-\phi/2)$ (using the half-angle identity).

So $\displaystyle\arctan\left(\frac{\sqrt{2}-u}{\sqrt{2-u^2}}\right)=\frac{\pi}{4}-\frac{\phi}{2}.$

The outer integral on T_2, with $du=\sqrt{2}\cos\phi\,d\phi$:

$$I_{T_2} = \int_{\pi/4}^{\pi/2}\frac{2}{\sqrt{2}\cos\phi}\left(\frac{\pi}{4}-\frac{\phi}{2}\right)\sqrt{2}\cos\phi\,d\phi = 2\int_{\pi/4}^{\pi/2}\left(\frac{\pi}{4}-\frac{\phi}{2}\right)d\phi.$$

$$= 2\left[\frac{\pi}{4}\phi-\frac{\phi^2}{4}\right]_{\pi/4}^{\pi/2} = 2\left[\left(\frac{\pi^2}{8}-\frac{\pi^2}{16}\right)-\left(\frac{\pi^2}{16}-\frac{\pi^2}{64}\right)\right] = 2\left[\frac{\pi^2}{16}-\frac{\pi^2}{16}+\frac{\pi^2}{64}\right] = \frac{\pi^2}{32}.$$

Therefore:

$$\sum_{n=1}^{\infty}\frac{1}{n^2} = 4(I_{T_1}+I_{T_2}) = 4\left(\frac{\pi^2}{16}+\frac{\pi^2}{32}\right) = 4\cdot\frac{3\pi^2}{32} = \frac{3\pi^2}{8}.$$

This is still not $\pi^2/6$. We must recheck the symmetry $v\mapsto -v$ factor. The integrand $2/(2-u^2+v^2)$ is even in v, so the integral over the full rotated square (which is symmetric in

v) equals $2\times$ the integral over $v \geq 0$. The full square in (u, v) has $v \in [-u, u]$ for $u \leq 1/\sqrt{2}$ and $v \in [-(\sqrt{2} - u), \sqrt{2} - u]$ for $u \geq 1/\sqrt{2}$. So the factor from $v \to -v$ is already accounted for within $T_1 \cup T_2$ (each triangle covers $v \in [0, \cdot]$ and its mirror). Hence *no extra factor of 2* from the v symmetry — the factor 4 above is wrong; it should be 2:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 2(I_{T_1} + I_{T_2}) = 2\left(\frac{\pi^2}{16} + \frac{\pi^2}{32}\right) = \frac{\pi^2}{8} + \frac{\pi^2}{16} = \frac{3\pi^2}{16}.$$

Still not $\pi^2/6$. There is an error somewhere in the domain decomposition. Let us restart from scratch with a clean, unambiguous decomposition following Apostol's presentation.

Clean derivation (following Apostol).

The rotation $x = (u + v)/\sqrt{2}$, $y = (u - v)/\sqrt{2}$ maps the unit square $[0, 1]^2$ to the rotated square with vertices $(0, 0)$, $(1/\sqrt{2}, 1/\sqrt{2})$, $(\sqrt{2}, 0)$, $(1/\sqrt{2}, -1/\sqrt{2})$. Since the integrand is even in v, we integrate over $v \geq 0$ and double:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 2 \iint_{R^+} \frac{2\, du\, dv}{2 - u^2 + v^2}, \tag{1}$$

where $R^+ = \{(u, v) : u \geq 0,\ v \geq 0,\ u + v \leq \sqrt{2},\ u \geq v\}$ (upper-half of the rotated square, i.e. $v \in [0, u]$, $u + v \leq \sqrt{2}$).

R^+ again splits at $u = 1/\sqrt{2}$: $T_1^+ = \{v \in [0, u],\ u \in [0, 1/\sqrt{2}]\}$ and $T_2^+ = \{v \in [0, \sqrt{2} - u],\ u \in [1/\sqrt{2}, \sqrt{2}]\}$.

From above: $I_{T_1^+} = \pi^2/16$ and $I_{T_2^+} = \pi^2/32$.

Substituting into (1):

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 2 \cdot 2(I_{T_1^+} + I_{T_2^+}) = 4\left(\frac{\pi^2}{16} + \frac{\pi^2}{32}\right) = \frac{\pi^2}{4} + \frac{\pi^2}{8} = \frac{3\pi^2}{8}.$$

This is incorrect. The standard proof gives $\pi^2/6$. Our error must be in claiming the $v \to -v$ symmetry gives a factor of 2: in fact the rotated square has $|v| \leq u$ for $u \leq 1/\sqrt{2}$ and $|v| \leq \sqrt{2} - u$ for $u \geq 1/\sqrt{2}$, so integrating $v \geq 0$ and doubling is correct and gives a factor of exactly 2 applied to $\iint_{R^+} 2/(2 - u^2 + v^2)\, du\, dv$. So we need:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 2(I_{T_1^+} + I_{T_2^+}) = 2\left(\frac{\pi^2}{16} + \frac{\pi^2}{32}\right) = \frac{\pi^2}{8} + \frac{\pi^2}{16} = \frac{3\pi^2}{16}.$$

This is still wrong. Let us recheck $I_{T_1^+}$.

$$I_{T_1^+} = \int_0^{1/\sqrt{2}} \int_0^u \frac{2\, dv\, du}{2 - u^2 + v^2}.$$

Inner integral: $\dfrac{2}{\sqrt{2 - u^2}} \arctan\left(\dfrac{u}{\sqrt{2 - u^2}}\right)$.

With $u = \sqrt{2} \sin\phi$, $\phi \in [0, \pi/4]$: $\sqrt{2 - u^2} = \sqrt{2}\cos\phi$, $u/\sqrt{2 - u^2} = \tan\phi$, $du = \sqrt{2}\cos\phi\, d\phi$:

$$I_{T_1^+} = \int_0^{\pi/4} \frac{2}{\sqrt{2}\cos\phi} \cdot \phi \cdot \sqrt{2}\cos\phi\, d\phi = 2\int_0^{\pi/4} \phi\, d\phi = 2 \cdot \frac{(\pi/4)^2}{2} = \frac{\pi^2}{16}.$$

This is correct.

Let us recheck $I_{T_2^+}$ using a cleaner substitution. With $u = \sqrt{2}\sin\phi$, $\phi \in [\pi/4, \pi/2]$: $\sqrt{2} - u = \sqrt{2}(1 - \sin\phi)$, $\sqrt{2 - u^2} = \sqrt{2}\cos\phi$, $du = \sqrt{2}\cos\phi\, d\phi$.

Inner integral upper limit: $\sqrt{2} - u = \sqrt{2}(1 - \sin\phi)$.

$$\int_0^{\sqrt{2}(1-\sin\phi)} \frac{2\,dv}{2 - u^2 + v^2} = \frac{2}{\sqrt{2}\cos\phi} \arctan\left(\frac{\sqrt{2}(1 - \sin\phi)}{\sqrt{2}\cos\phi}\right) = \frac{\sqrt{2}}{\cos\phi} \arctan\left(\frac{1 - \sin\phi}{\cos\phi}\right).$$

Using the identity $\dfrac{1 - \sin\phi}{\cos\phi} = \tan\left(\dfrac{\pi}{4} - \dfrac{\phi}{2}\right)$:

$$= \frac{\sqrt{2}}{\cos\phi}\left(\frac{\pi}{4} - \frac{\phi}{2}\right).$$

Outer integral:

$$I_{T_2^+} = \int_{\pi/4}^{\pi/2} \frac{\sqrt{2}}{\cos\phi}\left(\frac{\pi}{4} - \frac{\phi}{2}\right)\sqrt{2}\cos\phi\, d\phi = 2\int_{\pi/4}^{\pi/2}\left(\frac{\pi}{4} - \frac{\phi}{2}\right) d\phi.$$

Let $\psi = \pi/2 - \phi$ (so $\phi = \pi/2 - \psi$, $d\phi = -d\psi$, limits $\pi/4 \to 0$):

$$= 2\int_0^{\pi/4}\left(\frac{\pi}{4} - \frac{\pi/2 - \psi}{2}\right) d\psi = 2\int_0^{\pi/4} \frac{\psi}{2}\, d\psi = \int_0^{\pi/4} \psi\, d\psi = \frac{(\pi/4)^2}{2} = \frac{\pi^2}{32}.$$

So indeed $I_{T_2^+} = \pi^2/32$ and:

$$I_{T_1^+} + I_{T_2^+} = \frac{\pi^2}{16} + \frac{\pi^2}{32} = \frac{3\pi^2}{32}.$$

The total integral over the rotated square (using v-symmetry):

$$\iint_{\text{square}} \frac{2\,du\,dv}{2 - u^2 + v^2} = 2(I_{T_1^+} + I_{T_2^+}) = \frac{3\pi^2}{16}.$$

But we need this to equal $\sum 1/n^2$. There is a discrepancy with $\pi^2/6$: $3\pi^2/16 \neq \pi^2/6$. We must have an error in the rotation step. Let us recheck.

The rotation sends $(x, y) \to (u, v)$ with $x = (u + v)/\sqrt{2}$, $y = (u - v)/\sqrt{2}$. Then $xy = \frac{u^2 - v^2}{2}$ and $1 - xy = 1 - \frac{u^2 - v^2}{2} = \frac{2 - u^2 + v^2}{2}$. So the integrand transforms as:

$$\frac{1}{1 - xy} = \frac{2}{2 - u^2 + v^2}.$$

The Jacobian of the inverse map $(u, v) \to (x, y)$ is 1 (orthogonal transformation), so $dx\,dy = du\,dv$. This is all correct.

The rotated unit square has corners: $(x, y) = (0, 0) \to (u, v) = (0, 0)$; $(x, y) = (1, 0) \to (u, v) = (1/\sqrt{2}, -1/\sqrt{2})$; $(x, y) = (0, 1) \to (u, v) = (1/\sqrt{2}, 1/\sqrt{2})$; $(x, y) = (1, 1) \to (u, v) = (\sqrt{2}, 0)$.

So the rotated square in (u, v) is indeed the rhombus with those four vertices. By v-symmetry, the integral over the upper half ($v \geq 0$) equals half the total. The upper half is:

$$\{u \geq v \geq 0\} \cap \{u \leq \sqrt{2} - v \text{ or } u \leq 1/\sqrt{2}\}.$$

More precisely, the upper half of the rhombus is the triangle with vertices $(0,0)$, $(1/\sqrt{2}, 1/\sqrt{2})$, $(\sqrt{2}, 0)$, bounded by $v \le u$ and $v \le \sqrt{2} - u$. This is $R^+ = \{v \in [0, \min(u, \sqrt{2}-u)]\}$.

For $u \in [0, 1/\sqrt{2}]$: $\min(u, \sqrt{2}-u) = u$, so $v \in [0, u]$ (T_1^+). For $u \in [1/\sqrt{2}, \sqrt{2}]$: $\min(u, \sqrt{2} - u) = \sqrt{2} - u$, so $v \in [0, \sqrt{2} - u]$ (T_2^+). This matches what we had.

The total integral is:

$$\iint_{\text{rhombus}} \frac{2}{2 - u^2 + v^2}\, du\, dv = 2 \iint_{R^+} \frac{2}{2 - u^2 + v^2}\, du\, dv = 4(I_{T_1^+} + I_{T_2^+}) = \frac{3\pi^2}{8}.$$

We are getting $3\pi^2/8 = 3\pi^2/8 \approx 3.70$, but $\pi^2/6 \approx 1.645$. Clearly $3\pi^2/8 \neq \pi^2/6$. This means we have an algebraic error somewhere. Let us numerically check $\sum_{n=1}^{1000} 1/n^2 \approx 1.6439$ and compute the double integral numerically.

$\iint_{(0,1)^2} \frac{dx\, dy}{1-xy}$: near $(1,1)$ the integrand diverges. The integral is improper but convergent: $\int_0^1 \int_0^1 (xy)^n dx\, dy = 1/(n+1)^2$ and $\sum_{n=0}^{\infty} 1/(n+1)^2 = \pi^2/6$. So $\pi^2/6$ is correct for the double integral.

The discrepancy must be in $I_{T_1^+}$. Let us recheck numerically. $I_{T_1^+} = 2\int_0^{\pi/4} \phi\, d\phi = 2 \cdot \pi^2/32 = \pi^2/16 \approx 0.617$. $I_{T_2^+} = \pi^2/32 \approx 0.308$. $4(I_{T_1^+} + I_{T_2^+}) = 4(\pi^2/16 + \pi^2/32) = 4 \cdot 3\pi^2/32 = 3\pi^2/8 \approx 3.70$. But $\pi^2/6 \approx 1.645$. Factor of ≈ 2.25 off.

The issue: the rhombus has $v \in [-u, u]$, not just $v \in [0, u]$, and similarly for T_2. So when we say "upper half" and double, we must be integrating v from $-\min(u, \sqrt{2}-u)$ to $+\min(u, \sqrt{2}-u)$, which gives exactly twice the $v \ge 0$ part. So a factor of 2 from v-symmetry is correct.

Let us try a direct numerical estimate of $I_{T_1^+}$: At $u = 0.5$: inner integral $= \int_0^{0.5} \frac{2\, dv}{2 - 0.25 + v^2} = \int_0^{0.5} \frac{2\, dv}{1.75 + v^2} = \frac{2}{\sqrt{1.75}} \arctan(0.5/\sqrt{1.75}) \approx \frac{2}{1.3229} \arctan(0.3780) \approx 1.5119 \cdot 0.3614 \approx 0.546$. Outer integral $\approx \int_0^{0.707} 0.546\, du \approx 0.386$. True $I_{T_1^+} = \pi^2/16 \approx 0.617$. The discrepancy suggests the inner integral formula is wrong.

Recheck: $\int_0^u \frac{2\, dv}{2 - u^2 + v^2}$. At $u = 0.5$: $2 - u^2 = 1.75$, upper limit $v = 0.5$, so $\frac{2}{\sqrt{1.75}} \arctan(0.5/\sqrt{1.75}) \approx 0.546$. OK.

So the outer integral is $\int_0^{1/\sqrt{2}} f(u)\, du$ where $f(u) = \frac{2}{\sqrt{2-u^2}} \arctan(u/\sqrt{2-u^2})$.

Substitution $u = \sqrt{2}\sin\phi$, $du = \sqrt{2}\cos\phi\, d\phi$, $\sqrt{2-u^2} = \sqrt{2}\cos\phi$, limit $u = 1/\sqrt{2} \Leftrightarrow \sin\phi = 1/2 \Leftrightarrow \phi = \pi/6$ (not $\pi/4$!).

We had an error in the upper limit. $u = 1/\sqrt{2}$ and $u = \sqrt{2}\sin\phi$ gives $\sin\phi = \frac{1/\sqrt{2}}{\sqrt{2}} = \frac{1}{2}$, so $\phi = \pi/6$, *not* $\pi/4$.

Correcting:

$$I_{T_1^+} = 2\int_0^{\pi/6} \phi\, d\phi = 2 \cdot \frac{(\pi/6)^2}{2} = \frac{\pi^2}{36}.$$

Now $I_{T_2^+}$: on T_2^+, $u \in [1/\sqrt{2}, \sqrt{2}]$ corresponds to $\phi \in [\pi/6, \pi/2]$.

$$I_{T_2^+} = 2\int_{\pi/6}^{\pi/2} \left(\frac{\pi}{4} - \frac{\phi}{2}\right) d\phi.$$

Let $\psi = \pi/2 - \phi$, $d\phi = -d\psi$, limits $\pi/3 \to 0$:

$$= 2\int_0^{\pi/3} \frac{\psi}{2}\, d\psi = \int_0^{\pi/3} \psi\, d\psi = \frac{(\pi/3)^2}{2} = \frac{\pi^2}{18}.$$

Total:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 4(I_{T_1^+} + I_{T_2^+}) = 4\left(\frac{\pi^2}{36} + \frac{\pi^2}{18}\right) = 4 \cdot \frac{\pi^2}{12} = \frac{\pi^2}{3}.$$

Still not $\pi^2/6$. We are off by a factor of 2.

The remaining error: the v-symmetry factor. When we say "upper half and double", we get: $\iint_{\text{rhombus}} = 2\iint_{R^+}$. And $\iint_{R^+} = I_{T_1^+} + I_{T_2^+}$ (integrating $2/(2-u^2+v^2)$ over $v \in [0, \min(u, \sqrt{2}-u)]$). So total $= 2(I_{T_1^+} + I_{T_2^+})$, not $4(I_{T_1^+} + I_{T_2^+})$.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 2\left(\frac{\pi^2}{36} + \frac{\pi^2}{18}\right) = 2 \cdot \frac{\pi^2}{12} = \frac{\pi^2}{6}. \quad \checkmark$$

Summary of the correct proof.

Step 1. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2} = \iint_{(0,1)^2} \frac{dx\,dy}{1-xy}.$

Step 2. Rotation $u = (x+y)/\sqrt{2}$, $v = (x-y)/\sqrt{2}$ gives $1 - xy = (2-u^2+v^2)/2$ and $dx\,dy = du\,dv$. The unit square maps to the rhombus R with vertices $(0,0)$, $(1/\sqrt{2}, \pm 1/\sqrt{2})$, $(\sqrt{2}, 0)$:

$$\iint_{(0,1)^2} \frac{dx\,dy}{1-xy} = \iint_R \frac{2\,du\,dv}{2-u^2+v^2}.$$

Step 3. By $v \mapsto -v$ symmetry, integrating over $R^+ = R \cap \{v \geq 0\}$ and doubling:

$$= 2\iint_{R^+} \frac{2\,du\,dv}{2-u^2+v^2} = 2(I_{T_1^+} + I_{T_2^+}).$$

Step 4. R^+ splits at $u = 1/\sqrt{2}$. On T_1^+: $u \in [0, 1/\sqrt{2}]$, $v \in [0, u]$. Using $u = \sqrt{2}\sin\phi$, $\phi \in [0, \pi/6]$:

$$I_{T_1^+} = \int_0^{1/\sqrt{2}} \frac{2}{\sqrt{2-u^2}} \arctan\left(\frac{u}{\sqrt{2-u^2}}\right) du = 2\int_0^{\pi/6} \phi\,d\phi = \frac{\pi^2}{36}.$$

Step 5. On T_2^+: $u \in [1/\sqrt{2}, \sqrt{2}]$, $v \in [0, \sqrt{2}-u]$. Using $u = \sqrt{2}\sin\phi$, $\phi \in [\pi/6, \pi/2]$, and $\arctan\left(\frac{\sqrt{2}-u}{\sqrt{2-u^2}}\right) = \frac{\pi}{4} - \frac{\phi}{2}$:

$$I_{T_2^+} = 2\int_{\pi/6}^{\pi/2} \left(\frac{\pi}{4} - \frac{\phi}{2}\right) d\phi = \int_0^{\pi/3} \psi\,d\psi = \frac{\pi^2}{18}.$$

Step 6.

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{n^2} = 2\left(\frac{\pi^2}{36} + \frac{\pi^2}{18}\right) = 2 \cdot \frac{3\pi^2}{36} = \frac{\pi^2}{6}.} \quad \square$$

Conclusion: $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$ $\square$

> *This result appears in the partition function of phonons at low temperature (Debye model), in the calculation of the Casimir effect, and in the study of the Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$, which encodes the distribution of prime numbers.*

4.3 Volume of the n-Dimensional Ball

Let $B_n(R) = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_1^2 + \cdots + x_n^2 \le R^2\}$. Define $V_n(R) = \mathrm{Vol}(B_n(R))$. By dimensional analysis, $V_n(R) = \omega_n R^n$ where $\omega_n = V_n(1)$.

Recursion. Slicing $B_n(R)$ at height $x_n = t$:

$$V_n(R) = \int_{-R}^{R} V_{n-1}\left(\sqrt{R^2 - t^2}\right) dt = \omega_{n-1} \int_{-R}^{R} (R^2 - t^2)^{(n-1)/2}\, dt.$$

With $t = R \sin\psi$: the integral evaluates via the Beta function $B(a, b) = \int_0^1 u^{a-1}(1 - u)^{b-1}\, du = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, giving the recursion:

$$\omega_n = \frac{\pi}{n/2}\, \omega_{n-2}, \qquad \omega_1 = 2, \quad \omega_2 = \pi.$$

Solving:

$$\boxed{V_n(R) = \frac{\pi^{n/2}}{\Gamma(n/2+1)}\, R^n.}$$

Specific values.

n	1	2	3	4	n
$V_n(R)$	$2R$	πR^2	$\frac{4}{3}\pi R^3$	$\frac{\pi^2}{2} R^4$	$\frac{\pi^{n/2}}{\Gamma(n/2+1)} R^n$

Note: $V_n(1) \to 0$ as $n \to \infty$ — the unit ball in high dimensions has vanishingly small volume relative to the unit cube. Almost all the volume of a high-dimensional ball is concentrated near its surface, a fact central to statistical mechanics and machine learning.

4.4 Green's Theorem, Stokes' Theorem, and the Divergence Theorem

We computed line integrals in Meeting 2 and found that for exact forms, $\oint_\gamma df = 0$. This is a special case of a much deeper result.

Key operators. For $\mathbf{F} = (F_1, F_2, F_3)$ and scalar f, we define:

$$\begin{aligned}
\text{Gradient:}\quad & \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) \\
\text{Divergence:}\quad & \nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \\
\text{Curl:}\quad & \nabla \times \mathbf{F} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) \\
\text{Laplacian:}\quad & \Delta f = \nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}
\end{aligned}$$

Green's theorem (2D). For a region $D \subseteq \mathbb{R}^2$ with smooth boundary ∂D:

$$\oint_{\partial D} P\, dx + Q\, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\, dy.$$

The left side is the line integral around the boundary; the right side integrates the "curl" of (P, Q) over the interior.

When $\partial Q/\partial x = \partial P/\partial y$ (exactness condition!), the right side is zero, recovering $\oint df = 0$ for exact forms. Green's theorem is the *explanation* of why exact forms integrate to zero on closed loops: the integrand of the area integral vanishes identically.

Classical Stokes' theorem (3D surface). For an oriented surface S with boundary curve ∂S:

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}.$$

A vector field $\mathbf{F}$ is conservative (i.e., $\mathbf{F} = \nabla f$) if and only if $\nabla \times \mathbf{F} = 0$. By Stokes, this means $\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed curve, exactly as we proved for exact forms.

Divergence theorem (Gauss's theorem). For a volume Ω with boundary surface $\partial\Omega$:

$$\iint_{\partial\Omega} \mathbf{F} \cdot d\mathbf{S} = \iiint_{\Omega} \nabla \cdot \mathbf{F}\, dV.$$

The flux of $\mathbf{F}$ through a closed surface equals the integral of its divergence over the enclosed volume.

Physical content.

- Gauss's law: $\nabla\cdot\mathbf{E} = \rho/\varepsilon_0$ (the non-exact form $\Omega = \mathbf{r}/r^3 \cdot d\mathbf{S}$ from Meeting 2 corresponds to $\nabla \cdot \mathbf{E} = 0$ away from charges, but the divergence theorem detects the charge at the origin).
- Faraday's law: $\nabla \times \mathbf{E} = -\partial\mathbf{B}/\partial t$ (a non-zero curl means the field is not conservative, i.e., the form $\mathbf{E} \cdot d\mathbf{r}$ is not exact in time-varying fields).
- Continuity equation: $\partial\rho/\partial t + \nabla \cdot \mathbf{J} = 0$ (the divergence theorem converts local conservation into a surface flux).

The unifying statement. All three theorems are special cases of a single formula on differential forms, known as the **generalised Stokes theorem**:

$$\int_{\partial M} \omega = \int_M d\omega.$$

Here M is any smooth manifold with boundary ∂M, and $d\omega$ is the *exterior derivative* of the form ω. In the language of Meeting 2: the exactness condition $d\omega = 0$ is precisely the condition $\nabla \times \mathbf{F} = 0$ (or $\partial P/\partial y = \partial Q/\partial x$ in 2D). The "measure of non-exactness" $d\omega$ is what gets integrated over the interior when we apply any version of Stokes' theorem.

Summary

Key formulas

Change of variables:	$\int_V f\,dx = \int_U f(\Phi(u))\,\lvert\det D\Phi\rvert\,du$
Polar area element:	$dx\,dy = r\,dr\,d\theta$
Spherical volume element:	$dx\,dy\,dz = r^2 \sin\theta\,dr\,d\theta\,d\varphi$
Gaussian:	$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$
Ball volume:	$V_n(R) = \frac{\pi^{n/2}}{\Gamma(n/2+1)} R^n$
Green's theorem:	$\oint_{\partial D} P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$

Looking ahead. The generalised Stokes theorem $\int_{\partial M} \omega = \int_M d\omega$ unifies everything we have done: line integrals of exact forms (Meeting 2), the topology of the punctured plane (winding number as a cohomological invariant), and the divergence and Stokes theorems. The machinery of differential forms and de Rham cohomology gives the complete answer to the guiding question of Meeting 2: a form ω integrates to zero over every closed surface of dimension k if and only if ω is exact, and the obstruction to exactness is measured by the k-th cohomology group $H^k(M)$.